

AI Powered Debt Relief & Financial Recovery Platform

Project Description:

The **AI Powered Debt Relief & Financial Recovery Platform** is an intelligent AI-powered web application developed to simplify and automate the debt management and financial recovery process for borrowers. The platform enables users to manage loan details, analyze financial health, generate AI-powered negotiation strategies, and receive adaptive settlement recommendations through a secure and modern digital environment.

Built using **React.js, FastAPI, Python, SQLite, SQLAlchemy ORM, and Google Gemini API**, the system provides a responsive, scalable, and borrower-friendly interface with intelligent financial processing capabilities and real-time AI-assisted decision support.

Scenarios

Scenario 1: AI-Powered Settlement Recommendation

A borrower adds loan details including outstanding amount, EMI, overdue duration, and monthly income. The system analyzes the financial profile and generates personalized settlement recommendations along with debt stress analysis and financial health insights.

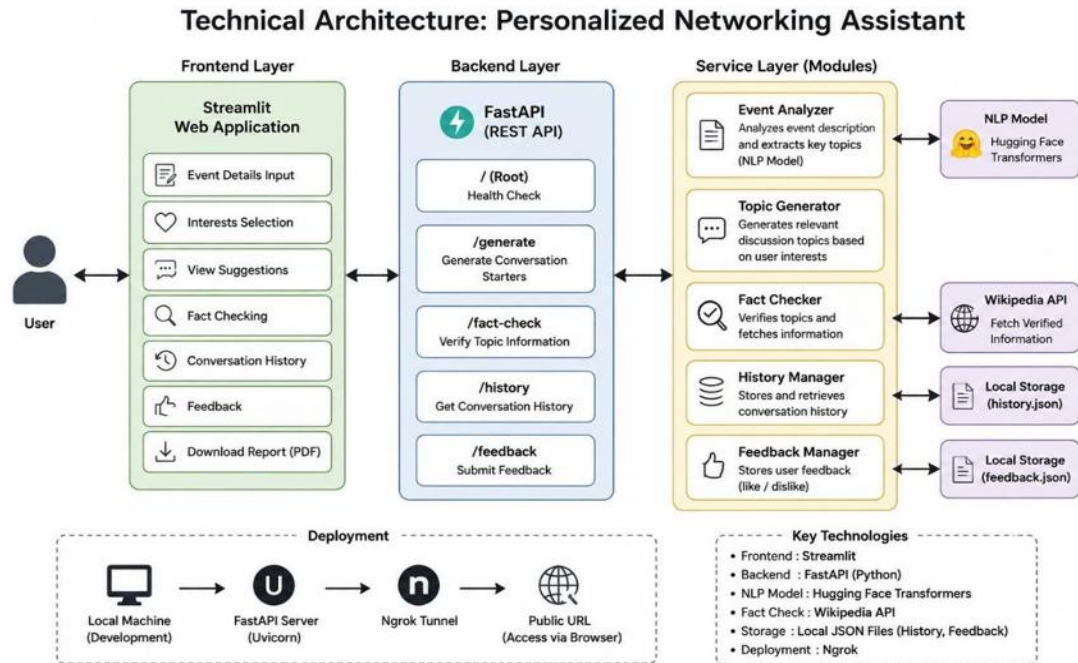
Scenario 2: Intelligent Negotiation Letter Generation

A borrower selects a loan account and requests a negotiation strategy. The platform uses Google Gemini AI to generate professional lender-specific negotiation emails and settlement request letters based on the borrower's financial condition and repayment capability.

Scenario 3: Financial Health Tracking & Loan Analysis

A borrower accesses the dashboard to monitor financial health metrics such as monthly surplus, EMI ratio, debt stress level, and settlement percentages. The platform also maintains AI-generated negotiation history and provides borrower guidance for improving debt management decisions.

Technical Architecture



Description:

AI Powered Debt Relief & Financial Recovery Platform follows a modular three-layer architecture.

- The **Frontend Layer** is developed using **React (Vite)** with CSS Modules to provide a responsive and user-friendly interface.
- The **Backend Layer** is developed using **FastAPI**, which manages API requests, loan processing, and financial calculations.
- The **Service Layer** contains reusable modules for financial analysis, Gemini AI integration, loan management, and recovery guidance.

The architecture is designed to be simple, modular, and easy to maintain.

Pre-requisites

- Basic Python Programming Knowledge
- React.js Fundamentals
- FastAPI Framework
- JavaScript, HTML, and CSS
- REST API Concepts
- SQLite Database Basics
- Google Gemini API
- Git and GitHub Basics
- Visual Studio Code
- Node.js and npm

Project Workflow

Milestone 1: Environment Setup and Architecture Design

Activity 1.1: Define Project Requirements

- Identify user requirements.
- Define the project objectives.
- Finalize the project scope.

Activity 1.2: Design System Architecture

- Create the system architecture.
- Define frontend and backend interaction.
- Plan AI integration.

Activity 1.3: Set Up Development Environment

- Install Python.
- Install Node.js.
- Install project dependencies.
- Configure the Gemini API key..

Activity 1.4: Create Project Structure

- Create folders for:
 - backend/
 - frontend/
 - routes/
 - models/
 - services/
 - components/
 - pages/

Milestone 2: Backend Development Activity 2.1:

Implement FastAPI Backend

Create API endpoints:

- /login
- /financial-analysis
- /loans
- /loans/user/{id}
- /loan-ai

Activity 2.2: Develop Loan Management Module

Functions:

- Add Loan
- View Loan
- Delete Loan

Responsibilities:

- Store loan details.
- Calculate total debt.
- Manage user loan records..

Activity 2.3: Develop Financial Analysis Module

Functions:

- Analyze financial condition.
- Calculate EMI.
- Calculate debt summary.

Responsibilities:

- Calculate financial risk.
- Generate simple financial status.
- Display debt overview..

Activity 2.4: Develop AI Integration Module

Integrate Google Gemini API.

Functions:

- Generate AI financial insights.
- Generate recovery suggestions.

Responsibilities:

- Analyze loan portfolio.
- Provide financial guidance.
- Handle API errors using fallback responses.

Milestone 3: Frontend Development Activity

Activity 3.1: Develop User Interface

Create pages for:

- Login
- Dashboard
- Add Loan
- Financial Analysis
- Recovery Matrix
- Guidance

Activity 3.2: Add User Features

Implement:

- Loan Management
- Financial Dashboard
- AI Analysis
- Recovery Suggestions
- Responsive Design

Milestone 4 Data Storage and Processing

Activity 4.1: Loan Data Storage

Store user loan information using:

- SQLite Database

- Browser Local Storage

Responsibilities:

- Save loan details.
 - Retrieve loan records.
 - Update dashboard data.
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Activity 4.2: Financial Analysis Processing

Responsibilities:

- Calculate debt summary.
- Generate financial reports.
- Display recovery information.

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 - Display recovery information.
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Milestone 5: Testing and Validation

Activity 5.1: Unit Testing

Create test file:

- test_system.py

Validate:

- Loan APIs
 - Financial calculations
 - AI response handling
 - Dashboard functionality
-

Activity 5.2: Swagger API Testing

Access Swagger UI:

<http://127.0.0.1:8000/docs>

Test:

- Login API
 - Loan APIs
 - Financial Analysis API
 - AI Integration API
-

Milestone 6: Deployment and Documentation

Activity 6.1: Local Deployment

Run FastAPI with Python:

```
python -m uvicorn backend.app.main:app --reload
```

```
Run React.js:
cd frontend
npm install
npm run dev
http://localhost:5173/
```

Activity 6.2: Version Control

Upload the project to GitHub.

Repository includes:

- Source Code
- Documentation
- Images
- Test Files
- README

Exploring the Application Pages

1. Login Page

Description:

The Login Page allows users to log in to the AI Powered Debt Relief & Financial Recovery Platform AI application using their User ID. After logging in, users can access their dashboard and use all the features of the application.

2. Dashboard

Description:

The Dashboard is the main page of the application. Users can add, view, and delete their loan details. It also shows important information such as Total Active Debt, Monthly EMI, and Settlement Savings to help users understand their financial situation.

3. Financial Analysis Page

Description:

The Financial Analysis page analyzes the user's loan details and shows their financial status. It displays the risk level as High Risk, Moderate Risk, or Healthy Buffer. It also provides simple AI-based suggestions to help users improve their financial health.

4. Recovery Matrix Page

Description:

The Recovery Matrix page compares the user's financial situation before and after using FinRelief AI. It provides simple recovery strategies and explains how better financial planning can help reduce debt.

5. Guidance & Compliance Page

Description:

The Guidance & Compliance page provides simple financial guidance and information about consumer protection rights. It helps users understand safe financial practices and their rights during loan repayment.

Conclusion

AI Powered Debt Relief & Financial Recovery Platform is an AI-powered web application that helps users manage their loans and understand their financial condition in a simple way. Users can add and manage loan details, view financial analysis, receive AI-based guidance, and learn about debt recovery through an easy-to-use interface.

The project is developed using React (Vite), FastAPI, SQLite, and the Google Gemini API. It also includes a backup response system that handles API errors, making the application more reliable. In the future, the project can be improved by adding more personalized financial analysis, loan tracking, cloud deployment, and support for more users.